Counterfactual Causality

Tutorial 6

Causality = Comparison

Core idea: X causes Y means changing X (holding everything else constant) changes Y.

For individual i:

- $Y_i(1)$ = outcome if treated ($X_i = 1$)
- $Y_i(0)$ = outcome if not treated ($X_i = 0$)
- Individual causal effect: $\tau_i = Y_i(1) - Y_i(0)$

The fundamental problem: We only observe ONE potential outcome per individual.

$$Y_i^{\text{obs}} = X_i \cdot Y_i(1) + (1 - X_i) \cdot Y_i(0)$$

Example: Coffee and Stress

Question: Does coffee increase stress? (Scale: 0-10)

Person	$Y_i(1)$ Drink	$Y_i(0)$ No Drink	τ_i	Observed
Alice	8	4	$+4$	8 (drinks)
Bob	7	6	$+1$	6 (no drink)
Carol	9	3	$+6$	9 (drinks)

Gray $=$ counterfactual (unobserved) Black $=$ factual (observed)

We can never observe both $Y_i(1)$ and $Y_i(0)$ for the same person!

From Individual to Average Effects

Can't observe $\tau_i = Y_i(1) - Y_i(0) \Rightarrow$ focus on averages.

Average Treatment Effect (ATE):

$$\text{ATE} = E[Y_i(1) - Y_i(0)] = E[Y_i(1)] - E[Y_i(0)]$$

Why equal? Linearity of expectation: $E[X - Y] = E[X] - E[Y]$

Using our example:

$$\text{Method 1:} \quad E[\tau_i] = \frac{4+1+6}{3} = 3.67$$

$$\text{Method 2:} \quad E[Y_i(1)] - E[Y_i(0)] = \frac{8+7+9}{3} - \frac{4+6+3}{3} = 8 - 4.33 = 3.67 \quad \checkmark$$

Naive Comparison Fails

Naive approach: Compare treated vs. control groups

$$E[Y_i \mid X_i = 1] - E[Y_i \mid X_i = 0]$$

Step 1: Apply observation rule
When $X_i = 1$, observe $Y_i(1)$; when $X_i = 0$, observe $Y_i(0)$:

$$= E[Y_i(1) \mid X_i = 1] - E[Y_i(0) \mid X_i = 0]$$

Step 2: Add and subtract $E[Y_i(0) \mid X_i = 1]$

$$= E[Y_i(1) \mid X_i = 1] - E[Y_i(0) \mid X_i = 1] + E[Y_i(0) \mid X_i = 1] - E[Y_i(0) \mid X_i = 0]$$

Selection Bias Decomposition

Step 3: Rearrange and simplify

$$E[Y_i \mid X_i = 1] - E[Y_i \mid X_i = 0]$$
$$= \underbrace{E[Y_i(1) - Y_i(0) \mid X_i = 1]}_{\text{True effect (ATT)}} + \underbrace{E[Y_i(0) \mid X_i = 1] - E[Y_i(0) \mid X_i = 0]}_{\text{Selection bias}}$$

Selection bias = difference between groups even without treatment

- $E[Y_i(0) \mid X_i = 1]$: treated group's counterfactual (unobserved)
- $E[Y_i(0) \mid X_i = 0]$: control group's actual outcome (observed)

Example: College Education

Question: Does college increase income?

People who go to college may be smarter, more motivated, wealthier families...
Even without college, they might earn more than non-college group.

Selection bias:

$$E[\text{Income without college} \mid \text{Went to college}] - E[\text{Income} \mid \text{No college}]$$
$$= \$50k - \$40k = \$10k$$

Suppose observed difference is \$40k:

$$\text{Observed} = \underbrace{\$30k}_{\text{True college effect}} + \underbrace{\$10k}_{\text{Selection bias}}$$

How to Eliminate Selection Bias?

Need: $E[Y_i(0) \mid X_i = 1] = E[Y_i(0) \mid X_i = 0]$ (and same for $Y_i(1)$)

Solution 1: Randomized Experiment (RCT)
Random assignment $\Rightarrow X_i \perp \{Y_i(0), Y_i(1)\}$
Selection bias $= 0$, simple comparison works!

Solution 2: Observational Methods

- Matching: control for confounders $\mathbf{Z}$
- Difference-in-Differences: use time variation
- Instrumental Variables: find exogenous variation
- Regression Discontinuity: use threshold discontinuities

Counterfactual Statements

Structure:

- **If clause** (subjunctive): states something false
- **Then clause** (conditional): what would happen

Example 1 (Past): Trump election and hate crimes

If clause: "If Trump had not won the 2016 election, ..."

Then clause: "... there would have been fewer hate crimes in 2017"

Example 2 (Future): Border wall

If clause: "If Trump built a border wall, ..."

Then clause: "... there would be fewer illegal immigrants"

Trump Example: Mathematical Form

Claim: Trump's election caused increased hate crimes in 2017

Let US be unit i:

- $X_i = 1$: Trump wins (factual, observed)
- $X_i = 0$: Trump loses (counterfactual)
- Y_i: hate crimes in 2017

Potential outcomes:

- $Y_i(1)$: hate crimes if Trump wins (observed)
- $Y_i(0)$: hate crimes if Trump loses (unobserved)

Causal effect: $\tau_i = Y_i(1) - Y_i(0)$
Problem: Only observe $Y_i(1)$. How to estimate $Y_i(0)$?

Not Counterfactual

Factual/Implicative: "if" is true
"If you heat water to 100°C, it boils."
Test: Can replace "if" with "when" $\Rightarrow$ not counterfactual

Predictive: "if" about possible future
"If it rains tomorrow, the game will be cancelled."
Both "rain" and "no rain" are possible futures.

Counterfactuals describe **hypothetical alternate timelines** that do not occur.

Practice Exercise

In groups (10 minutes):

1. Choose a causal claim (e.g., "Online education improves grades")
2. Define: What is $Y_i(1)$? $Y_i(0)$? τ_i?
3. Create a table with 3 hypothetical students:

Student	$Y_i(1)$	$Y_i(0)$	τ_i	Observed
?	?	?	?	?

4. Write as counterfactual: "If student had not taken online course, ..."

Summary

Key takeaways:

1. Causality = comparing same unit under different treatments
2. One potential outcome is factual, the other counterfactual
3. Fundamental problem: cannot observe counterfactuals
4. Simple comparison = True effect + Selection bias
5. Need methods to eliminate selection bias (RCT, matching, DID, IV, RDD)

Next lecture: Learn these methods in detail!

Questions?